



Electronics Manufacturing Jobs to Multiply by 2020

Here we explore the kind of job opportunities available in electronics industry in India, specially in the surface mount technology (SMT) sector. Also covered are skills required and pay packages that may be expected

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Electronics manufacturing industry in India needs to grow at a much faster pace due to the ever-growing demand for consumer electronics, IT and telecom goods. This demand is expected to grow to US\$ 400 billion by 2020, according to some estimates. In case manufacturing in India does not keep pace with the demand, country's import bill for electronics will exceed the import bill for crude oil.

Amarpreet Singh, manager-manufacturing, Centum Electronics says, "To support this demand and reduce the currency outflow from India, Indian government has come up with a very detailed industrial policy that encourages manufacturing industry in India. Indian manufacturing industry is growing and so are the skill requirements to support electronics industry."

Four broad fields for manufacturing

In electronics manufacturing value chain, there are four broad fields where electronics engineers fit in, namely, electronics tool development software industry, design organisations (who use these tools to develop electronics designs), original equipment manufacturers (OEMs, who use the designs and integrate them in their product systems

themselves or through electronics manufacturing services providers) and, finally, the qualification and failure analysis labs, according to Ankan Mitra, vice president, SMTA India Chapter.

Software tools industry for electronics design and manufacturing, and second, components, PCBs and system design houses are the two major fields where opportunities for engineers exist at present. Mitra says, "There is a huge industry for people with an electronics background, but it is aligned towards software tools development for electronics design industry. There is also a huge requirement for talented engineers in design companies."

Next come the OEMs, who use services of all these design companies to manufacture their products in their own factories or outsource it to electronic manufacturing services (EMS) companies. The fourth broad field to make a career in is failure analysis and qualification labs in India. Mitra informs, "In case of field failures, specialist engineers come into picture. Field engineers are specialised engineers who identify issues and report problems faced by customers and service engineers. They are in-charge of carrying out critical repairs on faulty machines and they play an important role in the electronics manufacturing field as well."

Apart from these, there are component suppliers or traders who exist across this value chain. Mitra says, "Engineers

procure components during the NPI cycle, manufacturing cycle and even during field failures. These are the five different layers, with the component supplier organisations being a parallel layer, which constitute different segments in the electronics manufacturing industry. Each segment provides different opportunities for engineers."

From design, production, test to SMT

The opportunities in electronics manufacturing industry are immense for fresh engineers as well as experienced professionals. "Entry-level roles vary widely depending on the company, products and geographies. But, usually, it involves assisting a team of mid-level managers who work on product development and manufacturing," informs Kaustubh Nande, manager-marketing, ANSYS Software (ANSYS India).

Sharing his views on opportunities in this field, T. Anand, principal consultant, Knewron says, "Opportunities exist in design, testing and assembly domain for engineers interested in these fields. However, entry-level roles in this domain focus on hands-on work rather than pure design or coding assignments. This means, one must understand and appreciate the importance of practical work experience before progressing in this field and making long-term career."

Surface mount technology (SMT) pro-